The examination is being carried out on the **following application documents**

Description, Pages

1-42 as originally filed

Claims, Numbers

1-15 as originally filed

Drawings, Sheets

1/8-8/8 as originally filed

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 KATIE Z LUO ET AL: "Augmenting Lane Perception and Topology Understanding with Standard Definition Navigation Maps", ARXIV.ORG, 7 November 2023 (2023-11-07), XP091655002, Retrieved from the Internet: URL:https://arxiv.org/pdf/2311.04079

D2 Fu Yanping ET AL: "TopoLogic: An Interpretable Pipeline for Lane Topology Reasoning on Driving Scenes", arXiv.org, 23 May 2024 (2024-05-23), XP093319960, Retrieved from the Internet: URL:https://arxiv.org/pdf/2405.14747

D3 Li Tianyu ET AL: "Graph-based Topology Reasoning for Driving Scenes", arXiv.org, 28 August 2023 (2023-08-28), XP093319963, Retrieved from the Internet: URL:https://arxiv.org/pdf/2304.05277

D4 WANG HENGLI ET AL: "End-to-End Interactive Prediction and Planning with Optical Flow Distillation for Autonomous Driving", 2021 IEEE/CVF CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION WORKSHOPS (CVPRW), IEEE, 19 June 2021 (2021-06-19), pages 2229-2238, XP033967760,

D5 ZHOU JIANG ET AL: "P-MapNet: Far-seeing Map Generator Enhanced by both SDMap and HDMap Priors", ARXIV.ORG, 15 March 2024 (2024-03-15), XP091701294, Retrieved from the Internet: URL:https://arxiv.org/pdf/2403.10521v1

D6 Sen Yang ET AL: "TopoSD: Topology-Enhanced Lane Segment Perception with SDMap Prior", arXiv.org, 22 November 2024 (2024-11-22), XP093319900, Retrieved from the Internet: URL:https://arxiv.org/pdf/2411.14751

1 It is to be noted that D1 refers explicitly to D3 as providing more detailed information on the lane-topology model (see D1 at section III.C, par. 1; section IV, par. 3; Table I), and therefore, the teaching of D3 is regarded as being incorporated into D1 (see EPC Guidelines G-IV.8). Similarly, D2 states explicitly that its geometric distance topology reasoning method is integrated with the SMERF model of D1 (see D2 at abstract; page 8, last paragraph; Table 5), and therefore, the teaching of D1 is regarded as being incorporated into D2.

2 **Article 84 EPC**

The application does not meet the requirements of Article 84 EPC for the reasons provided below.

2.1 In **claim 1**, the term "*onboard perception model*" would normally be understood as a perception model that is deployed onboard the vehicle. However, the description par. [0056] of the application gives a broader meaning to said term, not limiting the perception model to be deployed on the vehicle, which is not clear from the wording of the claim alone.

2.2 In **claim 3**, it is not clear what is the difference between the "*associated spatial feature*" and the "*associated-region lane position*" feature, as the lane position feature is also a spatial feature. The description par. [0088]-[0089], [0093]-[0095], [0103] and Figure 4 of the application appear to define different processing on a standard definition map for obtaining said features, but the difference in said feature extraction processes is not clearly reflected in the claim.

2.3 **Claims 9** and **11** have been drafted as separate independent claims of the same claim category without their subject-matter falling within one or more of the exceptional situations set out in Rule 43(2) EPC. The same applies to **claims 10** and **13**. Thus, the claims lack conciseness and do not satisfy the requirements of Article 84 EPC in combination with Rule 43(2) EPC.

3 Novelty and Inventive Step (Articles 52(1), 54 and 56 EPC)

3.1 The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of **claim 1** is not new within the meaning of Article 54(1) and (2) EPC.

3.1.1 Specifically, D1 discloses (the references in parentheses apply to said document):

A method for perceiving a road environment **(see section III, par. 1-2, the perceiving of lane centerlines and lane topology)**, *comprising:*
acquiring an associated-region lane attribute and an information to be detected, wherein the information to be detected is collected by an onboard sensor and represents a target region where a vehicle is traveling **(see section III, par. 1-2 and Figure 2, wherein a camera mounted on a vehicle acquires an image of the road on which the vehicle travels, with said image including information to be detected)**, *the associated-region lane attribute corresponds to an associated region, and the associated region and the target region meet a predetermined similarity condition* **(see sections III.A, III.B and Figure 2; for the same road area on which the vehicle travels, a local standard definition (SD) map is being used to extract lane attributes, including lane polylines and the respective road type)**; *and*
processing the associated-region lane attribute and the information to be detected by using an onboard perception model to obtain a road perception information of the target region **(see abstract, section III.C and Figure 2, wherein lane polylines/road types and multi-camera features are being processed to obtain lane topology prediction for the environment of the autonomous vehicle)**.

3.1.2 Thus, the subject-matter of **claim 1** lacks novelty over the disclosure of D1 (Article 54 EPC).

3.2 The additional technical features of **claims 2-6** do not involve an inventive step over the disclosure of D1 (Article 56 EPC) for the reasons provided below:

Claim 2: The neural network model of D1 includes an encoder and a decoder (see Figure 2); the encoder obtains features in a bird's eye view (BEV) space based on the multi-camera features and the associated SD map features, and provides said BEV features to the decoder of D3 for being processed by a cross-attention network module (see D1 at section III, par. 2, at sections III.C, IV.A, at Figure 2; see D3 at Figure 2); to fuse said BEV features with the readily available SD map features, which are also in BEV space, would be a trivial modification known in the art (see, for example, D5 at section IV.A and Figure 2).

Claims 3, 4: D1 extracts SD map features, including lane polyline positions and lane types, and applies cross-attention, including spatial cross-attention, in the fusion process with the multi-camera features to obtain BEV features (see sections III.B, III.C and Figure 2); the lane polyline positions define spatial information for the lanes, and therefore, they can be regarded as spatial features. It is also to be noted that convolutional neural networks are commonly used in the art for extracting spatial features from rasterised SD maps (see, for example, D5 at section IV.A and Figure 2).

Claim 5: D1 obtains an SD map-augmented feature map in a BEV space, and provides said BEV feature map to the lane-topology model decoder of D3 for predicting lane centerlines and affinity matrices (see D1 at section III, par. 2; section III.C; section IV.A). Said decoder includes a layer that processes lane query features "Q_l^{i-1}" fine-tuned by a previous layer of the decoder, as well as keys "K" and values "V" obtained by the BEV feature map, to obtain an intermediate fusion feature (see D3 at Figure 2), and processes said intermediate fusion feature with an adjacency matrix "A_{ll}^{i-1}" to update the lane queries and predict lane centerlines/affinity matrices (see D3 at sections 3.3.3 and 3.3.4). It is to be noted that D3 obtains the adjacency matrix "A_{ll}^{i-1}" by processing, with a topology classification network (topology head), the lane queries of a previous iteration (see section 3.3.3, par. 1 and 3.4.2; Figure 2). However, it would be an obvious alternative to determine said adjacency matrix by processing, with a topology classification network, the most updated lane queries, namely the lane queries that are provided as output by the centerline deformable decoder in the current iteration (see, for example, D2 at section 3 and Figure 2).

Claim 6: D1 uses multi-camera images to predict lane centerlines and an affinity matrix for the association of said centerlines (see section III.C and Figure 2); it would be obvious for the skilled person to include in said process input from additional modalities, e.g. LiDAR point clouds, as multi-modal processing is commonly used in the art for improving the robustness of a detection method (see, for example, D5 at section III, par. 1 and Figure 2).

3.3 The subject-matter of **claims 7-8** is directed to a training method of the perception model used in **claim 1**, with said training including a knowledge distillation mechanism with a teacher model using a feature loss and a road perception information loss. Such training process with a teacher model is commonly used in the art (see, for example, D4 at section 3, especially sections 3.4-3.5, and Figures 2, 3), and therefore, **claims 7-8** do not involve an inventive step over the disclosure of D1 (Article 56 EPC).

3.4 The subject-matter of **claims 9-15** corresponds to the subject-matter of **claims 1-8**. Therefore, the objections raised above for **claims 1-8** apply *mutatis mutandis* to **claims 9-15.**

3.5 It is to be noted that D2 uses the SMERF model of D1 (see D2 at section 3, at page 8, last paragraph and at Table 5), and therefore, the objections raised above over the disclosure of D1 apply *mutatis mutandis* over the disclosure of D2. Moreover, the subject-matter of **claims 1, 7, 9-15** lacks novelty (Article 54 EPC) over the disclosure of D4 (see said document at section 3, especially sections 3.4-3.5, and Figures 2, 3), while the subject-matter of **claims 1, 6, 9, 11-12, 14-15** lacks novelty (Article 54 EPC) over the disclosure of D5 (see said document at sections III, IV and at Figure 2).

4 Further remarks

From description par. [0102]-[0110], [0131]-[0134] and Figures 4, 5, it is understood that the application uses a specific encoder and decoder for perceiving the road environment which can be reflected in **claim 5**, **when dependent on claim 4**, after certain clarifications (see item 2 above). If the

Applicant decides to include said subject-matter in **claim 1**, it is requested to indicate what would be the significance of the new claim and why it would not be obvious over the cited prior art.